

Cryptography Basics

Name: Matyas Abraham

IDNO: CTC-207-26

Overview

This room covers the fundamentals of cryptography and symmetric encryption — why encryption matters in the first place, the vocabulary around plaintext and ciphertext, a run through a few historical ciphers, the difference between symmetric and asymmetric encryption, and the small amount of math (modulo and XOR) that keeps showing up underneath it. I went through all seven tasks and finished the room at 100%. Below is a walkthrough of what each task covered, my answers to the challenge questions, and screenshots as proof of work.



Room overview — all seven tasks completed.

Task 1 — Introduction

This task is just the room's setup — it introduces cryptography as the practice of securing information so only the intended recipient can read it, and lays out what the rest of the room will cover: why cryptography matters, the core vocabulary, some historical ciphers, symmetric vs. asymmetric encryption, and the basic math behind it.

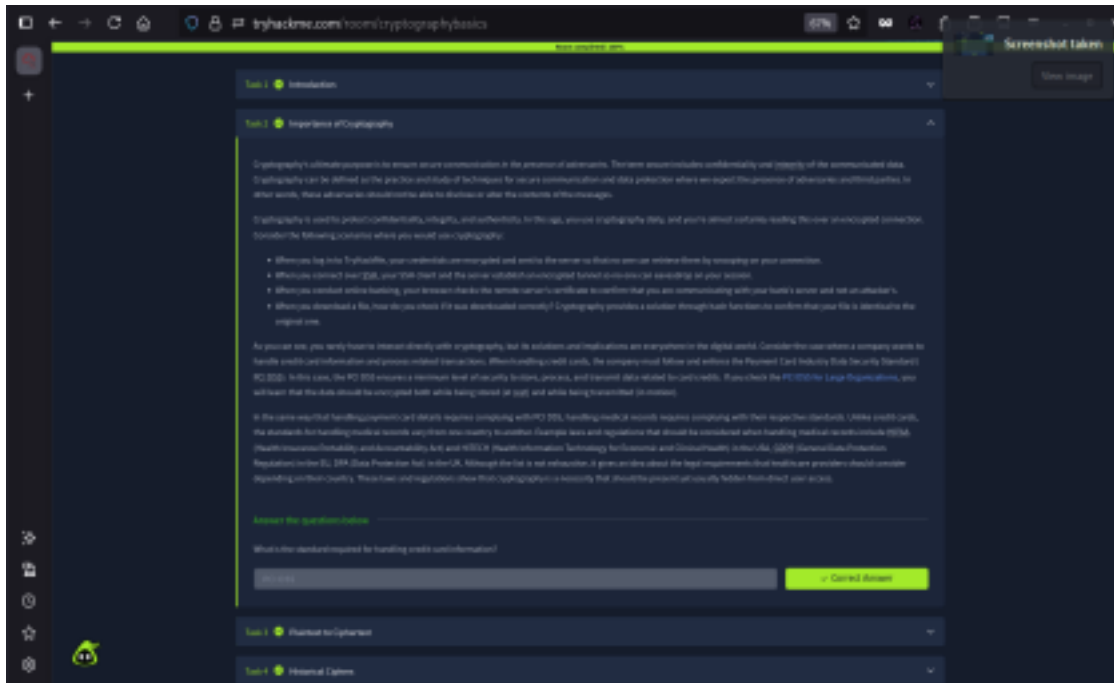
Task 2 — Importance of Cryptography

This task made the case for why cryptography isn't optional in most industries — it's often a legal requirement, not just a good idea. Handling medical records, for instance, is governed by different laws depending on the country: HIPAA and HITECH in the USA, GDPR in the EU, and DPA in the UK. None of these laws mandate encryption by name in every case, but they all expect sensitive data to be protected, and encryption is usually how that gets done in practice.

The task's question was about a different kind of sensitive data — payment information. The standard that applies there is PCI DSS (Payment Card Industry Data Security Standard), which sets the requirements for anyone storing, processing, or transmitting credit card data.

Q: What is the standard required for handling credit card information?

A: PCI DSS.



Task 2 — Importance of Cryptography

Task 3 — Plaintext to Ciphertext

This task covered the core vocabulary: plaintext is the original, readable data before anything is done to it. Encryption is the process of converting that plaintext into ciphertext using a cipher and a key — the cipher can be public knowledge, but without the key, recovering the plaintext should be practically impossible. Decryption is just the reverse: turning ciphertext back into plaintext, again using the cipher and the (correct) key.

Q: What do you call the encrypted plaintext?

A: Ciphertext.

Q: What do you call the process that returns the plaintext?

A: Decryption.



Task 3 — Plaintext to Ciphertext

Task 4 — Historical Ciphers

This task walked through the Caesar cipher as the classic starting example — each letter in the plaintext gets shifted by a fixed number of positions in the alphabet, and the same shift in reverse decrypts it. It also mentioned a few other historical ciphers worth knowing about: the Vigenère cipher from the 16th century, the Enigma machine from World War II, and the one-time pad from the Cold War. There's also a link to the Cryptii website, which lets you experiment with Caesar shifts interactively and watch the text transform in real time.

The challenge question gave the ciphertext `XRPCTCRGNEI` and said it was encrypted with a Caesar cipher. I worked through the possible shifts (25 of them, since shifting by 26 just gets back to the original) until the output actually read as English, which turned out to be `ICANENCRYPT`. It's a good illustration of why the Caesar cipher is considered broken by modern standards — with only 25 possible shifts, brute-forcing it by hand takes a couple of minutes at most.

Q: Knowing that `XRPCTCRGNEI` was encrypted using Caesar Cipher, what is the original plaintext? A: `ICANENCRYPT`.



Task 4 — Historical Ciphers

Task 5 — Types of Encryption

This task introduced Alice and Bob, the fictional pair used in basically every cryptography example to represent two parties trying to communicate securely, and drew the line between the two main types of encryption. Symmetric encryption uses the same key for both encryption and decryption, which means that key has to stay secret and never get shared with anyone outside the conversation. Asymmetric encryption uses two different keys instead — a public key that anyone can use to encrypt a message, and a private key, kept secret, that's the only thing that can decrypt it.

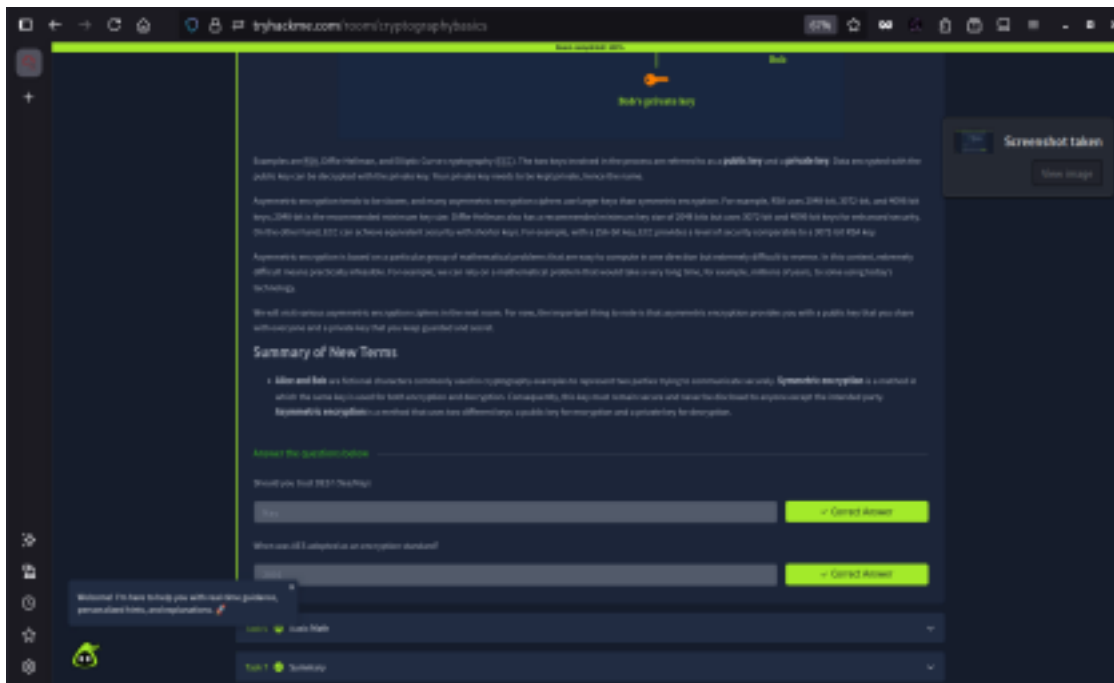
The task also touched on two symmetric algorithms specifically: DES and AES. DES uses a 56-bit key, which was considered secure decades ago but is now short enough to be brute-forced without much difficulty, which is why it's no longer trusted for anything serious. AES replaced it and was formally adopted as the encryption standard in 2001, and it's still what's used today.

Q: Should you trust DES? (Yea/Nay)

A: Nay — its 56-bit key is short enough to brute-force with modern hardware.

Q: When was AES adopted as an encryption standard?

A: 2001.



Task 5 — Types of Encryption

Task 6 — Basic Math

This task covered the two math operations that show up constantly once you get into how ciphers actually work under the hood: XOR and modulo.

XOR ($\oplus$) compares two binary numbers bit by bit and returns 1 where the bits differ and 0 where they match. Modulo ($\%$) returns the remainder after dividing one number by another — so $23\%6$ is 5, because 23 divided by 6 is 3 with a remainder of 5. One thing worth remembering about modulo is that it isn't reversible: given only the result of $x\%5 = 4$, there are infinitely many values of x that could satisfy that, so you can't work backward to a single answer.

Q: What's $1001\ 1010\oplus$

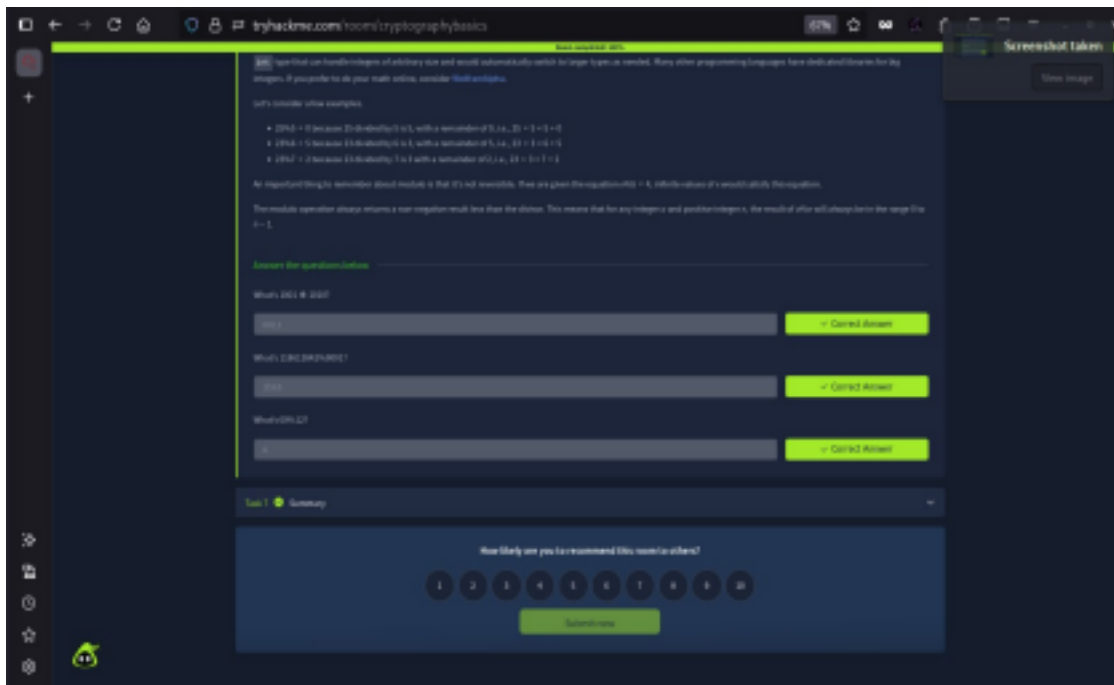
A: 0011 — comparing bit by bit: $1\ 1=0, 0\ 0=0, 0\ 1=1, 1\ 0=1$. $\oplus\ \oplus\ \oplus\ \oplus$

Q: What's $118613842\%9091$?

A: 3565.

Q: What's $60\%12$?

A: 0 — 60 divides evenly by 12, so there's no remainder.



Task 6 — Basic Math

Task 7 — Summary

The room wrapped up by tying the pieces together: cryptography is what keeps data confidential, both at rest and in transit, and it's frequently a legal requirement rather than just good practice. Plaintext becomes ciphertext through encryption, and only the right key can reverse that through decryption. Historical ciphers like Caesar and Vigenère show how the field has evolved, and today's encryption splits into symmetric (one shared key, fast, but the key has to be protected) and asymmetric (a public/private key pair, which solves the key-sharing problem but is heavier computationally). Underneath most of it, especially symmetric encryption, is basic math — XOR and modulo being the two that came up here.

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Room completed 100%

Task 1 Introduction

Task 2 Importance of cryptography

Task 3 Plaintext to Ciphertext

Task 4 Historical Ciphers

Task 5 Types of Encryption

Task 6 Basic Math

Task 7 Summary

In this room, we learned about the importance of cryptography and some of the problems that it solves. We also introduced symmetric and asymmetric encryption systems. Finally, we explained the Diffie-Hellman key exchange, the RSA algorithm, and the SHA-256 hash function. In the next room, Public Key Cryptography Basics, we will see how asymmetric cryptography and we how they solve the problems we face in the digital world.

Answer the questions below

Before proceeding to the next room, make sure you have taken notes of all the topics and concepts introduced in this room.

Do you understand?

Correct Answer

How likely are you to recommend this room to others?

12345678910

Submit rating